

## Inventory Management System

### System Process and Functional Description

#### 1. System Overview

The **Inventory Management System** is designed to manage inventory resources within five engineering departments of a faculty. The system controls inventory requests, approvals, stock tracking, maintenance handling, and annual reporting.

The system manages two main inventory types:

- **Capital Inventory**
- **Simple Inventory**

The system also manages **inventory lifecycle events**, including:

- Requesting new inventory
- Approval workflows
- Maintenance and repair handling
- Removal of unusable inventory
- Stock tracking
- Annual reporting

The system supports **role-based access control**, where each role has specific responsibilities and permissions.

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#### 2. Departments

The system supports **five engineering departments**:

1. Civil Engineering
2. Computer Engineering
3. Mechanical Engineering
4. Electrical and Electronic Engineering
5. Interdisciplinary Engineering

Each department has its own users:

- 1 Head of Department (HOD)
- 1 Management Assistant (MA)

- Lab In-Charges
- Lab Technical Officers (Lab TO)

Additionally, there is a central role:

- Inventory Management Officer (IMO)
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### **3. User Roles and Responsibilities**

#### **3.1 Inventory Management Officer (IMO)**

The Inventory Management Officer has **administrative authority over the inventory system.**

##### **Responsibilities**

- Manage inventory items
- Manage stock levels
- Approve capital item requests
- Generate official documents for welfare team approval
- Monitor inventory status across departments
- Generate reports
- Configure stock thresholds
- Send notifications

##### **Use Cases**

1. Login / Logout
2. Manage Inventory Items
  - Add new inventory
  - Update inventory details
  - Delete inventory records
3. Categorize inventory
4. View department-wise inventory
5. Generate reports
6. Set stock threshold alerts
7. Approve or reject capital requests
8. Generate approval document for welfare team

9. Send notifications
  10. Monitor system inventory usage
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### **3.2 Head of Department (HOD)**

The HOD supervises inventory activities within their department.

#### **Responsibilities**

- Approve capital inventory requests
- Review department inventory
- Provide remarks for approvals
- Monitor department inventory usage

#### **Use Cases**

1. Login / Logout
  2. View department inventory
  3. Approve or reject capital inventory requests
  4. Add remarks for requests
  5. Track request progress
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### **3.3 Lab In-Charge**

Lab In-Charge is responsible for **reviewing technical requirements of lab equipment requests**.

#### **Responsibilities**

- Review lab equipment requests
- Approve or reject lab requests
- Forward approved requests to HOD

#### **Use Cases**

1. Login / Logout
2. View lab inventory
3. Approve or reject lab equipment requests
4. Add technical remarks
5. Send approved requests to HOD

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### 3.4 Management Assistant (MA)

The Management Assistant manages **purchasing and inventory updates**, especially for furniture and general items.

#### Responsibilities

- Process approved requests
- Record purchase details
- Update stock levels
- Notify departments about completed purchases

#### Use Cases

1. Login / Logout
2. View approved requests
3. Record purchase information
4. Update inventory stock (Stock-In)
5. Send notifications to relevant staff

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### 3.5 Lab Technical Officer (Lab TO)

Lab TO manages day-to-day lab inventory operations.

#### Responsibilities

- Request lab equipment
- Issue equipment to users
- Record returns
- Track maintenance needs

#### Use Cases

1. Login / Logout
2. View lab inventory
3. Create capital item request
4. Issue item
5. Record item return

6. Create maintenance log
  7. View system alerts
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### 3.6 System Admin

System Admin manages **user accounts and system configuration**.

#### Responsibilities

- Manage system users
- Assign roles
- Configure system security
- Monitor system activity

#### Use Cases

1. Login / Logout
  2. Create user profiles
  3. Update user profiles
  4. Delete users
  5. Assign roles and permissions
  6. Configure system security
  7. Monitor system health
  8. View audit logs
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## 4. Inventory Categories

All inventory items belong to **categories** and contain detailed **specifications**.

### Inventory Types

#### 4.1 Capital Inventory

Capital inventory consists of major items such as:

- Furniture
- Laboratory equipment
- Laboratory components
- Large stationary assets

These items usually require **multi-level approval**.

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#### 4.2 Simple Inventory

Simple inventory consists of small consumable items such as:

- Dust cloths
- Minor maintenance materials
- Small daily-use items

These require **simpler approval processes**.

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### 5. Inventory Data Structure

Each inventory item contains:

- Item ID
  - Item Name
  - Category
  - Inventory Type (Capital / Simple)
  - Department
  - Specifications
  - Quantity
  - Stock threshold
  - Status
  - Purchase details
  - Maintenance history
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### 6. Inventory Request Workflows

#### 6.1 Simple Inventory Request Workflow

Simple inventory requests follow a **short approval process**.

##### Process

1. Lab TO submits request
2. Lab In-Charge reviews request

3. If approved → forwarded to MA
4. MA reviews request
5. If approved → process completed
6. Inventory stock updated

### **Workflow**

Lab TO → Lab In-Charge → MA → Completed

If any role rejects the request → request is marked **Rejected**.

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## **6.2 Capital Inventory Request Workflow**

Capital items require **multiple approvals**.

### **Process**

1. Lab TO creates request
2. Lab In-Charge reviews request
3. If approved → forwarded to HOD
4. HOD reviews request
5. If approved → forwarded to IMO
6. IMO reviews request
7. IMO generates official approval document
8. Document is sent to Welfare Team (external system)
9. Welfare Team reviews request outside the system
10. Approved document is returned
11. Document uploaded to system
12. Request completed

### **Workflow**

Lab TO  
→ Lab In-Charge  
→ HOD  
→ IMO  
→ Welfare Team (External)  
→ Document Upload  
→ Completed

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## 7. Request Status Tracking

The system must track the **current progress of all requests**.

Example statuses:

- Submitted
- Pending In-Charge Approval
- Pending HOD Approval
- Pending IMO Approval
- Sent to Welfare Team
- Welfare Approved
- Completed
- Rejected

Users must always see **current request status**.

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## 8. Maintenance and Repair Handling

If equipment becomes damaged:

### Repair Process

1. Lab TO creates maintenance log
2. Item sent to Welfare Team
3. Welfare team evaluates item

Possible outcomes:

### Repairable

- Item repaired
- Returned to department
- System updated with repair details

### Not Repairable

- Welfare team replaces with new component
  - Inventory updated
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## 9. Removal of Unusable Inventory

If inventory becomes unusable:

Examples:

- Broken equipment
- Damaged components
- Old furniture

### Process

1. Request for removal sent to Welfare Team
  2. Welfare team evaluates item
  3. If approved → item marked **Disused**
  4. System updates inventory status
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## 10. Annual Evaluation Process

At the end of the year:

The Welfare Team visits faculty premises.

They evaluate:

- Broken furniture
- Old equipment
- Repairable items
- Items that must be removed

The system records:

- Items repaired
  - Items replaced
  - Items removed from usability
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## 11. Notification System

The system must provide notifications for:

- Request approvals
- Request rejections

- Stock updates
- Low stock alerts
- Maintenance alerts

If inventory quantity drops below **threshold level**, system shows alerts.

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## **12. Inventory Threshold Monitoring**

IMO sets minimum stock levels.

When quantity falls below threshold:

- System generates notification
  - Relevant users are alerted
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## **13. Inventory Reporting**

The system generates reports including:

### **Annual Inventory Report**

- Total inventory items
- Department-wise inventory
- Capital inventory summary
- Simple inventory summary

### **Removal Report**

- Number of unusable items removed
- Reason for removal

### **Maintenance Report**

- Repaired items
- Replaced items
- Maintenance history

### **Stock Report**

- Current stock levels
  - Low stock alerts
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## 14. Audit and Security

The system must maintain **audit logs** including:

- User logins
- Inventory changes
- Request approvals
- Request rejections
- Stock updates

Security features include:

- Password policies
- Role-based permissions
- Session timeout
- Activity logging

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## 15. Key System Modules

1. User Management Module
2. Inventory Management Module
3. Request Management Module
4. Approval Workflow Module
5. Maintenance Management Module
6. Notification Module
7. Reporting Module
8. Security and Audit Module